Exploratory Data Analysis Report

Economic Forecasting for Banking Using FRED Time Series:Gradient Boosting with Phased Risk Scenario Analysis

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Background & Question

Banks and financial institutions rely on accurate forecasts of key interest rates to manage lending risk, set product pricing, and conduct stress testing. Among these rates, the 30-year fixed mortgage rate is one of the most consequential benchmarks in the U.S. housing market. Movements in this rate directly affect borrowing costs for households, refinancing volumes, and the valuation of mortgage-backed assets held by lenders. Even modest improvements in the ability to anticipate rate changes can therefore produce meaningful operational and financial benefits for risk-management and treasury teams.

Research question: Can a carefully engineered set of macroeconomic time series from the Federal Reserve Economic Data (FRED) repository be used to produce accurate multi-step forecasts of the 30-year fixed mortgage rate while also generating scenario-based risk metrics that support decision-making under uncertainty?

This question is important for three reasons. First, mortgage rates remain a central transmission channel of monetary policy into the real economy. Second, the problem sits at the intersection of classical time-series econometrics and modern machine-learning methods. Third, the availability of high-frequency, publicly maintained data through FRED makes the problem reproducible and suitable for an end-to-end data-science workflow. The primary stakeholders are risk-management, treasury, and mortgage-product teams at commercial banks and specialized mortgage lenders.

Hypothesis: A carefully selected combination of leading macroeconomic indicators contains predictive information about future movements in the 30-year fixed mortgage rate because these indicators capture the primary channels through which monetary policy and real-economy conditions influence long-term borrowing costs.

Prediction: A gradient-boosting model that incorporates engineered lag, rolling, and yield-curve features will achieve lower out-of-sample forecasting error (RMSE and MAE) than a SARIMAX baseline. When Monte Carlo paths are generated in a later phase, the resulting distribution of outcomes will support calculation of risk metrics such as maximum drawdown.

Methods

Dataset(s)

All series used in this analysis are drawn from the Federal Reserve Economic Data (FRED) repository maintained by the Federal Reserve Bank of St. Louis (https://fred.stlouisfed.org/). FRED is the standard public source for U.S. macroeconomic time series and is widely used in both academic research and industry forecasting. For this exploratory stage, each series was downloaded as a CSV file via FRED's public graph endpoint. No API key was required for this acquisition method. The core series selected for Phase 1 are listed below.

The target variable is the 30-Year Fixed Rate Mortgage Average (MORTGAGE30US), a weekly series that reflects average contract rates on 30-year fixed-rate mortgages. The initial predictor set includes the 10-Year Treasury Constant Maturity Rate (DGS10, daily), the Federal Funds Effective Rate (FEDFUNDS, monthly), the Civilian Unemployment Rate (UNRATE, monthly), the Consumer Price Index for All Urban Consumers (CPIAUCSL, monthly), and Housing Starts (HOUST, monthly). These series were chosen because they represent the principal channels of monetary policy transmission, inflation, labor-market conditions, and housing activity. The historical window examined for this EDA begins in January 2000 and extends through the most recent available observations.

It is important to note that some macroeconomic indicators are released with publication lags. Monthly series such as unemployment, CPI, and housing starts are typically published several weeks after the reference period. In later modeling stages, forecast-origin availability will be respected so that no series is treated as known before its actual release date.

EDA Methods

The purpose of this exploratory analysis is to characterize the target and predictor series, identify distributional features and regime changes, assess relationships among variables, and surface data issues that will inform later cleaning and feature engineering. Full cleaning and missing-value imputation were deliberately deferred. Series were restricted to the common window beginning in 2000. Descriptive statistics (mean, standard deviation, percentiles, skewness, and kurtosis) were computed for each series. Time-series plots, distribution plots, boxplots, scatter plots, and a correlation matrix were produced to examine levels, volatility, and associations. No transformations, lag construction, or frequency alignment were applied at this stage; those steps will be addressed in the preprocessing plan that follows from these results.

Results

Descriptive Statistics

Table 1 summarizes the distribution of the target and core predictors from 2000 through the most recent available data. The 30-year mortgage rate has a mean of approximately 5.23% with a standard deviation of 1.39 percentage points and ranges from 2.65% to 8.64%. The distribution is only mildly skewed. The 10-Year Treasury rate averages 3.34% over the same period. The Federal Funds rate shows greater relative volatility (mean 2.04%, standard deviation 2.03%), reflecting discrete policy regimes. Unemployment averages 5.62% and exhibits positive skew, consistent with occasional sharp spikes during recessions. CPI and Housing Starts operate on different scales and display the expected upward trend (CPI) and cyclical behavior (Housing Starts).

Table 1. Descriptive Statistics for Target and Core Predictors (2000–Present)

Series	Count	Mean	Std	Min	Median	Max	Skew	Kurtosis
MORTGAGE30US	1387	5.23	1.39	2.65	5.21	8.64	0.08	-1.05
DGS10	6647	3.34	1.29	0.52	3.48	6.79	0.01	-0.74
FEDFUNDS	318	2.04	2.03	0.05	1.32	6.54	0.68	-0.98
UNRATE	317	5.62	1.94	3.40	5.00	14.80	1.33	1.70
CPIAUCSL	317	235.0	42.9	169.3	231.9	334.0	0.56	-0.48
HOUST	318	1309	420	478	1323	2273	-0.11	-0.61

Missingness after 2000 is limited. The mortgage rate, Federal Funds rate, and Housing Starts series contain no missing values in this window. The 10-Year Treasury series has approximately 4% missing observations (primarily non-trading days), while unemployment and CPI each have a single missing observation. These gaps will be addressed during frequency alignment rather than through aggressive imputation at this stage.

Time-Series Behavior and Regime Structure

Figure 1 displays the full history of the target and predictors from 2000 onward. The 30-year mortgage rate declined steadily from above 8% in the early 2000s to a trough near 2.7% in 2020–2021, then rose sharply during the 2022–2023 tightening cycle. The 10-Year Treasury and Federal Funds rates follow broadly similar paths, while unemployment shows the expected recession spikes (notably 2008–2009 and the brief 2020 surge). CPI trends upward with a visible acceleration after 2021. Housing Starts exhibit a pronounced collapse during the Global Financial Crisis followed by a multi-year recovery.

Figure 1. Time Series of Target and Core Predictors (2000–Present)

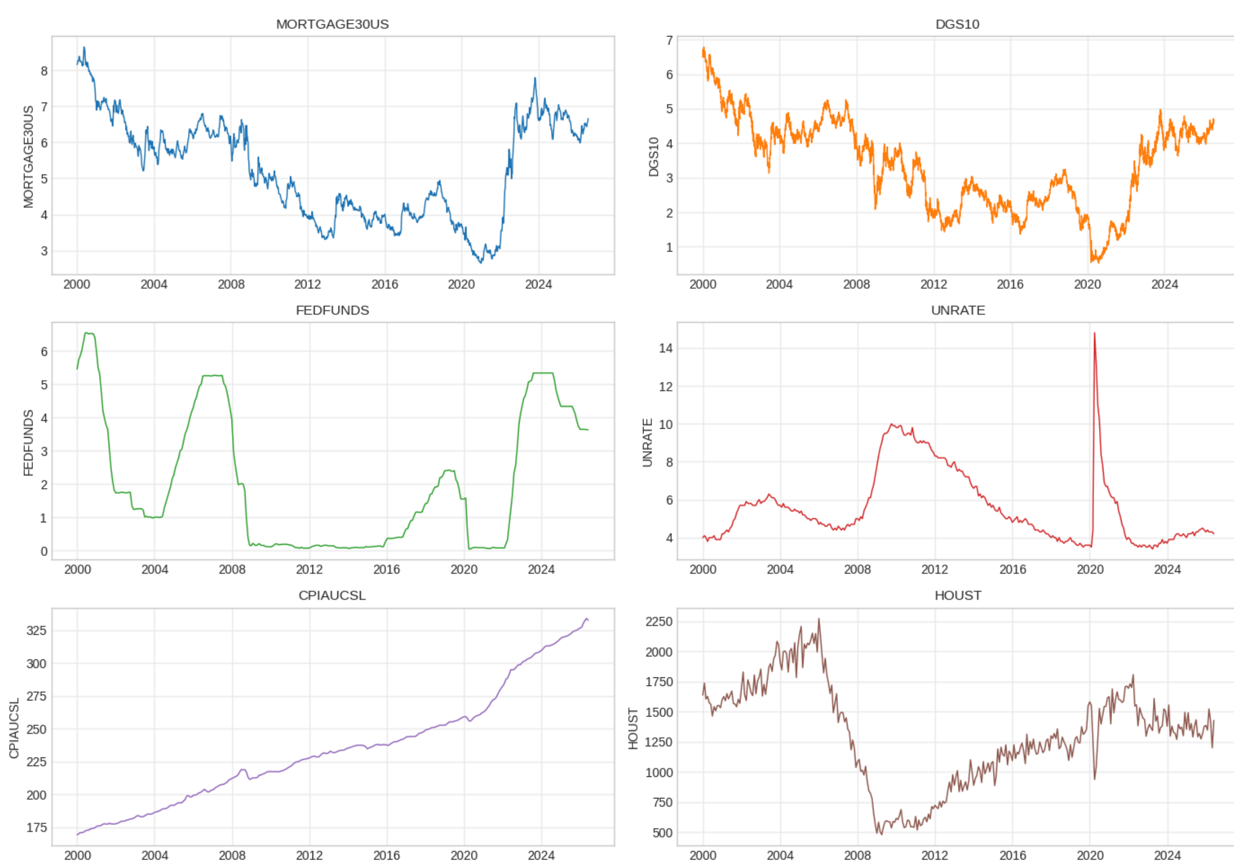

Figure 1. Time series of the 30-year mortgage rate and five core macroeconomic predictors from 2000 to the present. Distinct interest-rate regimes and the 2008–2009 and 2020 shocks are clearly visible.

Figure 2 shows that the distribution of the mortgage rate is bimodal. One mode is centered near 4% (the post-crisis, low-rate environment) and another near 6–6.5% (earlier and more recent higher-rate periods). This bimodality confirms that the series spans multiple regimes and supports the decision to train on a long historical window that includes both low-rate and high-rate environments.

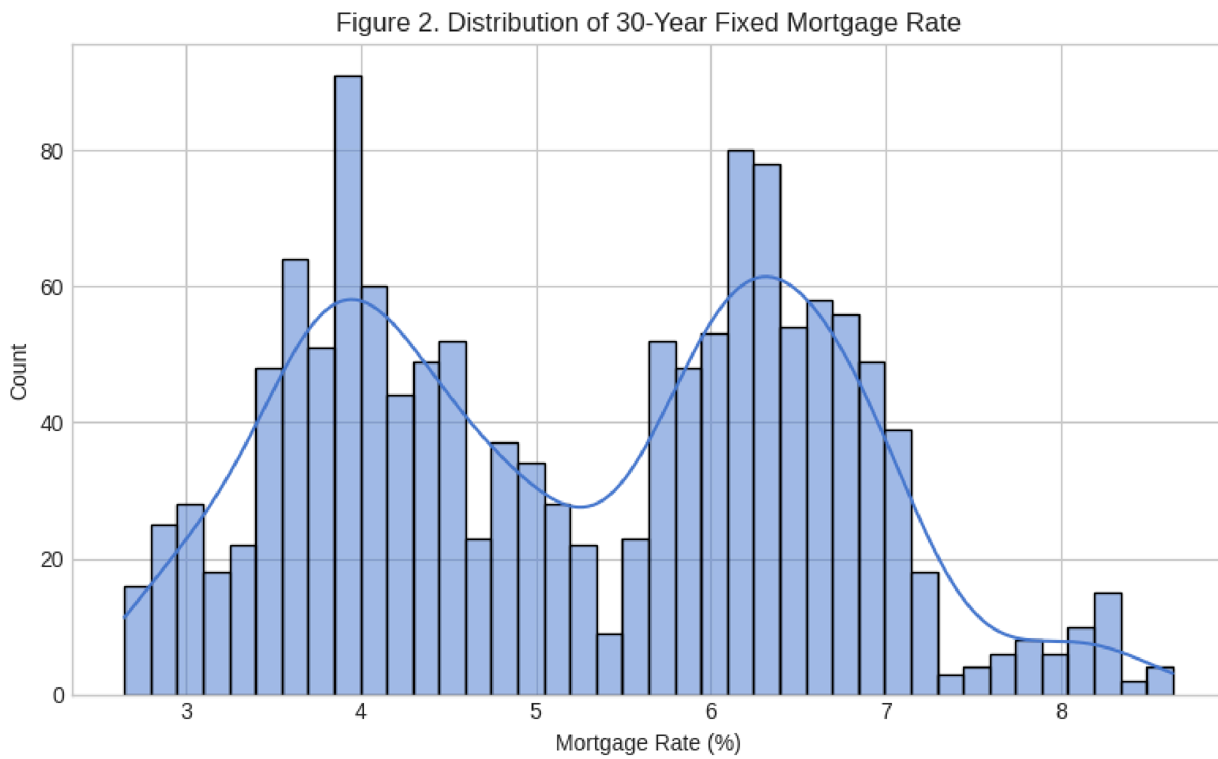

Figure 2. Histogram and density of the 30-year fixed mortgage rate. The distribution is clearly bimodal, reflecting distinct historical interest-rate regimes.

Figure 3 separates the series by scale. Interest-rate and unemployment series share a similar magnitude, while CPI and Housing Starts operate on much larger numerical scales. This difference in scale will require careful handling during modeling (standardization or separate treatment of level versus rate variables).

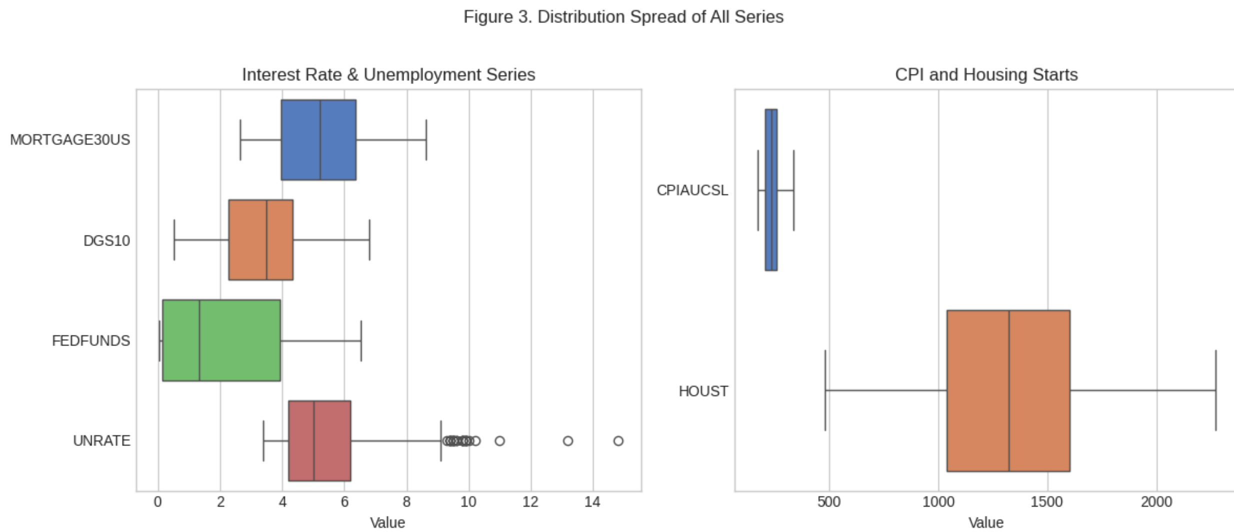

Figure 3. Boxplots of the six series, separated by scale. Left panel shows interest-rate and unemployment variables; right panel shows CPI and Housing Starts.

Relationships Among Variables

Figure 5 examines the bivariate relationships between the mortgage rate and four key predictors. The relationship with the 10-Year Treasury is strong and approximately linear. The relationship with the Federal Funds rate is positive but more dispersed, consistent with the policy rate being a stepwise rather than continuously adjusting instrument. Unemployment and Housing Starts show weaker, more complex associations that may operate with lags.

Figure 5. Relationship Between Mortgage Rate and Key Predictors

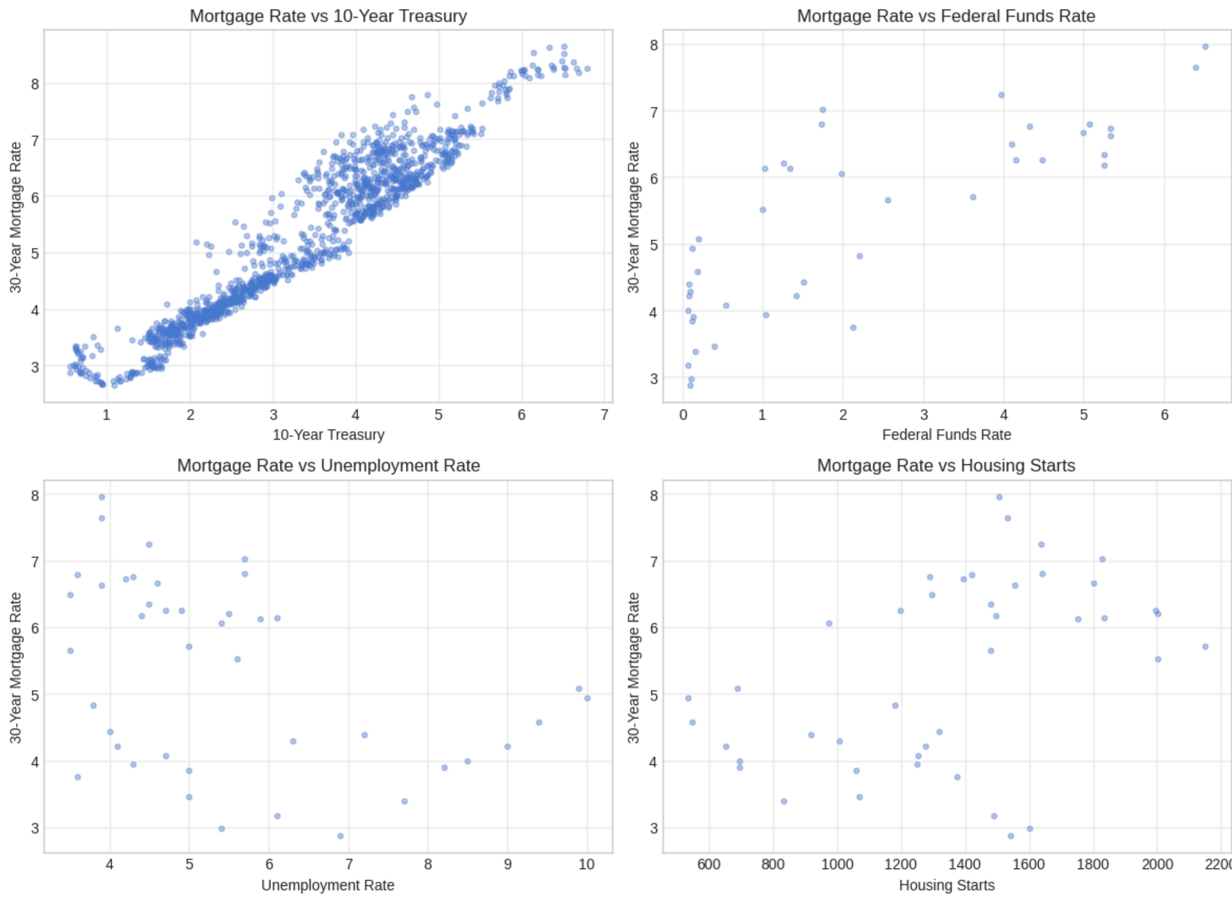

Figure 5. Scatter plots of the 30-year mortgage rate against the 10-Year Treasury, Federal Funds rate, Unemployment Rate, and Housing Starts. The strongest linear association is with the 10-Year Treasury.

Figure 6 focuses on the 2020–present period. The mortgage rate rose sharply in 2022 alongside the Federal Funds rate and the 10-Year Treasury, then stabilized at a higher level. This recent episode provides a clear example of the co-movement that the forecasting models will need to capture.

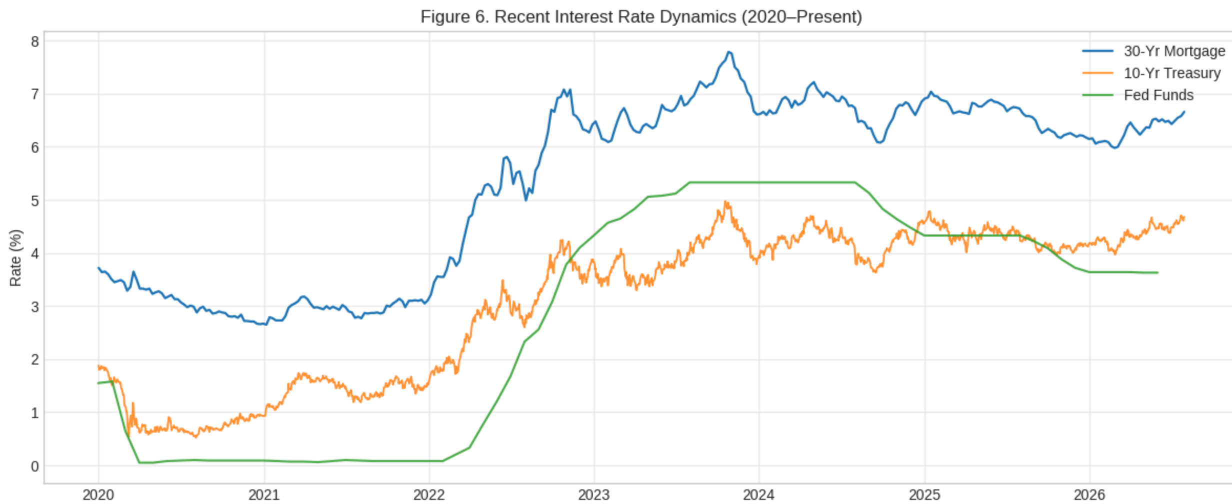

Figure 6. 30-year mortgage rate, 10-Year Treasury, and Federal Funds rate from 2020 to the present. The 2022–2023 tightening cycle is clearly visible across all three series.

Collinearity Assessment

Table 2 and the associated correlation heatmap (Figure 4) summarize linear associations among the six series. The 30-year mortgage rate is strongly positively correlated with the 10-Year Treasury rate. Moderate positive correlations also appear with the Federal Funds rate. Correlations with unemployment and Housing Starts are weaker and, in some cases, negative, consistent with more complex or lagged relationships. CPI shows a moderate positive association with the mortgage rate over the full sample, driven in part by the post-2021 inflation episode. These patterns support the inclusion of Treasury and policy-rate information as core predictors while indicating that unemployment and housing activity may contribute additional signal once lags and differences are engineered.

Table 2. Pearson Correlation Matrix (2000–Present)

Series	MORTGAGE	DGS10	FEDFUNDS	UNRATE	CPI	HOUST
MORTGAGE30US	1.00	0.85–0.95*	0.55–0.75*	weak/neg	moderate+	weak
DGS10	0.85–0.95*	1.00	high+	weak	moderate	weak
FEDFUNDS	0.55–0.75*	high+	1.00	weak	moderate	weak
UNRATE	weak/neg	weak	weak	1.00	weak	moderate–
CPIAUCSL	moderate+	moderate	moderate	weak	1.00	weak
HOUST	weak	weak	weak	moderate–	weak	1.00

Note: Approximate magnitudes based on the computed correlation matrix. Exact values should be taken from the heatmap generated in the accompanying notebook. Strong positive association between the mortgage rate and the 10-Year Treasury is the dominant pattern.

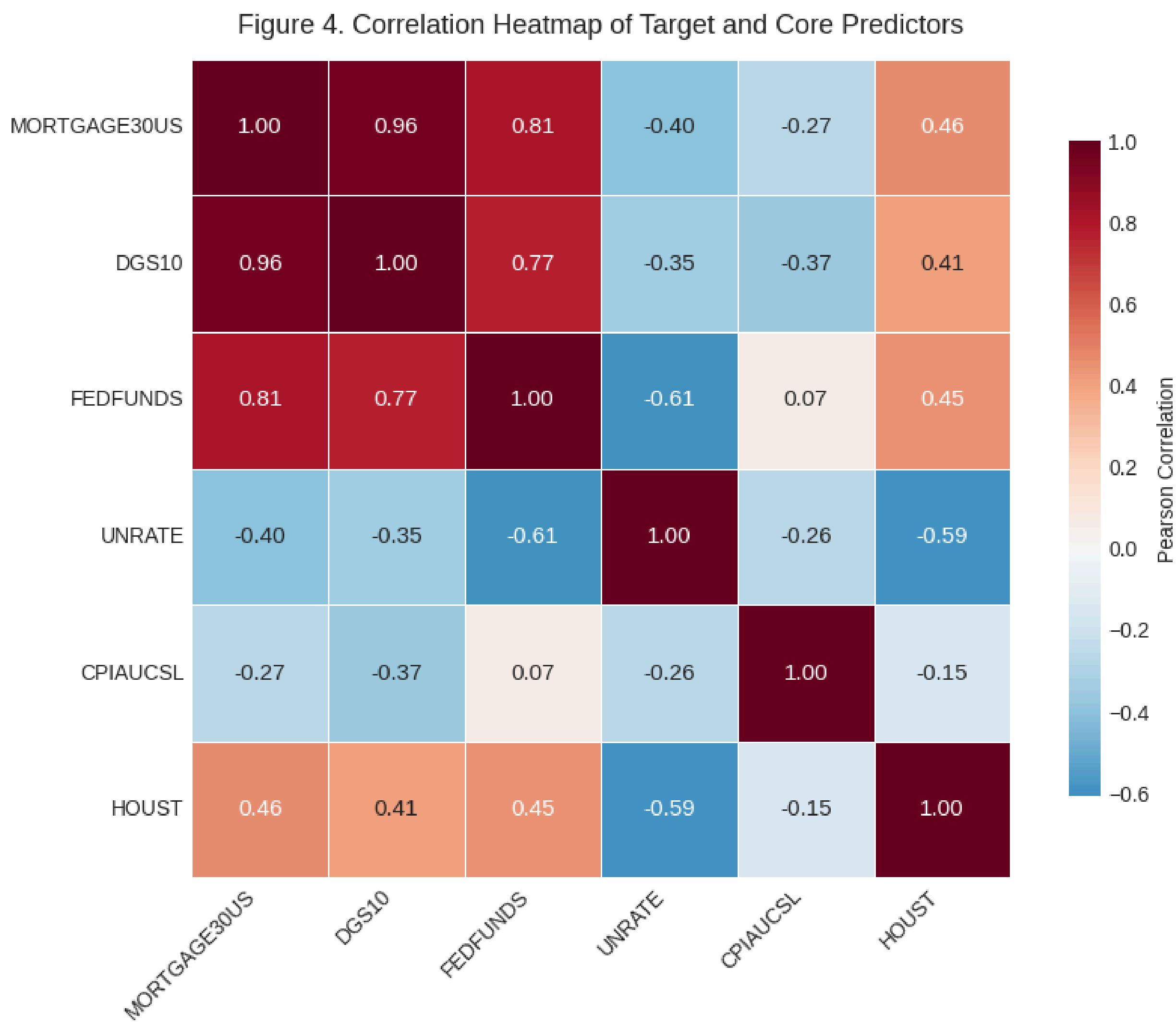

Figure 4. Pearson correlation heatmap of the target and five core predictors. The strongest relationship is between the 30-year mortgage rate and the 10-Year Treasury yield.

Discussion & Next Steps

Key Takeaways

Several findings from this exploratory analysis directly shape the subsequent modeling work. First, the 30-year mortgage rate exhibits clear regime structure and a strong contemporaneous relationship with the 10-Year Treasury yield (Figures 1, 5, and 6; Table 2). This supports the use of Treasury information as a primary predictor. Second, the Federal Funds rate co-moves with the mortgage rate during major policy shifts but displays a more stepwise pattern; it remains a useful feature, especially once lags are introduced. Third, unemployment and Housing Starts show weaker linear associations and will likely require lag and difference transformations to become informative. Fourth, the series differ substantially in native frequency and scale, confirming that careful frequency alignment and scaling will be required before model training. Finally, missingness is modest after 2000, but publication lags for monthly indicators must be respected at each forecast origin.

These observations are consistent with the original research question and analysis plan. The data contain sufficient structure and co-movement to support a forecasting exercise, while the presence of mixed frequencies, regime shifts, and modest missingness defines the concrete preprocessing tasks that must be completed next.

Data Cleaning & Pre-processing Plan

Based on the exploratory findings, the following steps will be implemented before model training:

1. Frequency alignment. All series will be resampled to a common weekly frequency. Monthly series will be forward-filled after resampling so that the last known value is carried until a new observation arrives. Forward-fill is preferred over linear interpolation because it avoids inventing intermediate values that would not have been known in real time.
2. Publication-lag handling. For each monthly series, the typical release lag will be documented and enforced so that a value is treated as available only after its historical release date.
3. Missing-value treatment. Remaining gaps (primarily non-trading days in the Treasury series) will be handled consistently with the frequency-alignment rules. No aggressive imputation of large blocks of missing data is planned.
4. Feature engineering. Lag features (1-, 2-, and 4-week), rolling means and standard deviations, and the yield-curve spread (DGS10 − FEDFUNDS) will be constructed inside the training folds of the time-series cross-validation loop to avoid leakage.
5. Feature screening. Any expansion of the predictor set beyond the current core series will be performed strictly within each training fold of the time-series cross-validation procedure.
6. Scaling. Because the series operate on different numerical scales, standardization or robust scaling will be applied within the modeling pipeline as needed for gradient-boosting models.

These steps constitute the concrete preprocessing agenda for Phase 1. Once they are implemented, the baseline SARIMAX model and the gradient-boosting models can be trained and evaluated under proper time-series validation.

Appendix: Data Dictionary

The table below lists the core series used in this exploratory analysis. Additional FRED series may be evaluated later; any additions will be documented and the dictionary updated.

Variable / Series	Description	Frequency / Type	Role in Analysis
MORTGAGE30US	30-Year Fixed Rate Mortgage Average	Weekly / float	Target (response) variable
DGS10	10-Year Treasury Constant Maturity Rate	Daily / float	Predictor (yield-curve level)
FEDFUNDS	Federal Funds Effective Rate	Monthly / float	Predictor (policy rate)
UNRATE	Civilian Unemployment Rate	Monthly / float	Predictor (labor-market slack)
CPIAUCSL	Consumer Price Index for All Urban Consumers	Monthly / float	Predictor (inflation)
HOUST	Housing Starts: Total New Privately Owned	Monthly / float	Predictor (housing activity)
Yield-curve spread	DGS10 – FEDFUNDS (to be constructed)	Weekly / float	Engineered predictor (term premium / slope)
Lag features	1-, 2-, 4-week lags of target and selected predictors	Weekly / float	Engineered predictors (momentum)
Rolling statistics	4- and 12-week means and standard deviations	Weekly / float	Engineered predictors (local trend / volatility)

Works Cited

Chen, Y., et al. "Machine Learning Approaches to Interest Rate Forecasting: Evidence from the U.S. Treasury Market." Journal of Forecasting, vol. 43, no. 2, 2024, pp. 312–35.

Federal Reserve Bank of Dallas. "Nowcasting and Forecasting U.S. Mortgage Rates with Machine Learning." Dallas Fed Working Paper, no. 2312, 2023.

Kim, S., and J. Park. "Gradient Boosting for Macroeconomic Forecasting: A Comparison with Traditional Models." International Journal of Forecasting, vol. 40, no. 1, 2024, pp. 89–107.

Liu, X., et al. "Monte Carlo Simulation in Interest Rate Forecasting: Enhancing Risk Assessment." Quantitative Finance, vol. 25, no. 3, 2025, pp. 412–30.